



Community Hero - Hyperlocal Problem Solver

Background

Communities frequently face issues such as potholes, water leakages, damaged streetlights, waste management concerns, and public infrastructure challenges. Reporting these issues is often fragmented, difficult to track, and lacks transparency.

Challenge

Build a platform that enables citizens to identify, report, validate, track, and resolve community issues through collaboration, data, and intelligent automation. The solution should encourage transparency, accountability, and community participation.

Example Features:

- Image and video-based issue reporting
- AI-powered issue categorization
- Geo-location and mapping
- Community verification
- Real-time issue tracking
- Impact dashboards
- Predictive insights
- Gamification for citizen engagement

Evaluation Focus

The solution should demonstrate how AI can help communities address local challenges more efficiently through improved reporting, verification, tracking, and resolution of issues.

Submission: JanSetu

1. Solution Overview

JanSetu shifts the civic grievance paradigm from standard 'passive complaints lists' to an **active, self-triage, and self-escalating operating system**. Instead of forcing citizens to navigate confusing municipal portals, select exact categories, or manually search for geolocations, JanSetu exposes a **Zero-UI voice interface** where citizens speak naturally in English, Hindi, or mixed dialects.

Behind the scenes, JanSetu orchestrates a **directed acyclic graph (DAG) of five specialized Gemini-powered agents** acting as autonomous municipal actors:

Agent	Core Reasoning & Job Description	API / Model
Triage Agent	Translates raw speech, categorizes context, extracts landmarks, and geocodes coords.	Gemini 1.5 Pro
Duplicate Detector	Groups spatial-temporal overlaps to prevent ticket duplication and spam.	Gemini 1.5 Pro
COI Engine	Formulates dynamic priority score based on severity, population density, and infrastructure risk.	Gemini 1.5 Pro
SLA Watchdog	Queries live web for nodal government officials and drafts formal escalation letters.	Gemini + Search Grounding
Verification Agent	Compares before/after resolution photos using visual reasoning to prevent municipal fraud.	Gemini Vision

2. Key Features

- **Zero-UI Voice Reporting:** Real-time transcriptions & translation of English, Hindi, or mixed voice notes.
- **Google Lens-Style Drag-and-Drop Dropzone:** Fluid drop zone supporting drag hover styling, upload feedback, size limits, and easy reset.
- **COI-Ranked Officer Dashboard:** Auto-sorts administrative list by cost-of-inaction value rather than standard chronology.
- **Interactive Heatmaps & Cluster Polygons:** Custom geocoded markers and risk overlay polygons showing spatial trends in real time.
- **Official Gov/NGO Nodal Directory:** Displays linked databases for CPGRAMS, MyGov, MoHUA sanitation services, Janaagraha, and Praja Foundation.
- **Immutable Transparency Ledger:** Chronological, public audit logs containing escalation documents and vision confirmation results.

3. Technologies Used

Core Web: React 18, Vite (HMR build orchestration), and React Router DOM.

Design System: Custom CSS using fluid layouts, dark-mode variables, dot-grid dynamic background, glassmorphism, and responsive columns.

Location Engine: Leaflet.js map layer + Web Geolocation API.

4. Google Technologies Utilized

- **Gemini 1.5 Pro:** Powers language translations, structural ticket parsing, and priority algorithms.
- **Gemini Search Grounding:** Performs queries on live government datasets to retrieve active engineer names and designations.
- **Gemini Vision:** Audits and verifies uploaded before/after resolution pictures.
- **Firebase Firestore:** Manages live tickets, ledger timeline entries, and cluster states.
- **Firebase Authentication:** Handles anonymous citizen sessions & email officer access.
- **Firebase Storage & Hosting:** Secures static file uploads under strict size parameters and hosts production app at `jansetu-dev.web.app`.